

REVISION 2026

Basic Surveying Lab

Diploma in Civil Engineering

Code: 2019

Credits: 1.5

Semester: 2

Course Outcomes (CO)

- **CO1:** Perform linear and angular measurements using chain and compass. (Apply)
- **CO2:** Determine level differences using simple and differential levelling. (Apply)
- **CO3:** Plot longitudinal profiles and prepare contour maps. (Apply)
- **CO4:** Measure horizontal angles and bearings using a theodolite. (Apply)

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

Module 1: Chain & Compass Survey

Expt 1: Equipment Introduction

Demonstration of basic survey equipment: Chain, Cross-staff, Plane table, and Compass.

Expt 3: Compass Survey

Determine inaccessible distances between points using compass and tape measurements.

Experiments, assignments and projects are provided to help students understand the concepts and applications of surveying.

Module 2: Levelling

Expt 4: Simple Levelling

Determine level difference between two points using a Dumpy level.

Expt 5: Rise and Fall Method

Determine reduced levels (RL) by the Rise and Fall method for a series of points.

Experiments, assignments and projects are provided to help students understand the concepts and applications of surveying. (RL) Reduced Levels are determined.

Module 3: Profile & Contouring

Expt 7: Profile Levelling

Plot longitudinal profiles and cross sections for the alignment of a road.

Expt 9: Contouring

Take spot levels and prepare a contour map for a given area.

பொருத்தமான தகவல்களைப் பெறும்படி கருத்துரைக்கப்பட்டுள்ள பகுதிகளைப் பார்வையிட்டு, அவற்றின் நிலைகளை எடுத்துக் கொள்ளவும். அவற்றின் அளவுகளைக் கவனமாகக் கவனித்து, அவற்றின் அளவுகளைப் பதிவு செய்யவும்.

Module 4: Theodolite Survey

Expt 10: Repetition Method

Measure horizontal angles with high precision using the repetition method with a theodolite.

Expt 12: Prolonging a Line

Prolong a straight line and measure the bearing of a line using a theodolite.

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Practice Model Question Paper

Practice Model Paper — not an official REV2026 paper

1. Determine the level difference between two given points using the Height of Instrument (HI) method.
2. Measure the horizontal angle between two objects using the repetition method with a theodolite.

Source Declaration

Curriculum Scheme: Revision 2026

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Content from other curriculum revisions was not merged.